

# Econ 101A

## Section 2

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We finish the math toolkit with the **Envelope Theorem**, then begin consumer theory: **preferences**, **utility functions**, and **indifference curves**. The link back to Section 1 is direct: the slope of an indifference curve comes straight from the Implicit Function Theorem.

## 1 The Envelope Theorem

Often we care about how the *optimal value* of an objective changes with a parameter, not how the optimal *choice* changes. The envelope theorem says the former is easy: only the direct effect of the parameter matters.

**Statement (unconstrained).** Let  $V(a) = \max_x f(x, a)$  with maximizer  $x^*(a)$ . Then

$$V'(a) = \left. \frac{\partial f}{\partial a} \right|_{x=x^*(a)}.$$

**Why?** By definition  $V(a) = f(x^*(a), a)$ . Differentiate using the chain rule:

$$V'(a) = \underbrace{f_x(x^*(a), a) \cdot x^{*'}(a)}_{=0 \text{ by FOC}} + f_a(x^*(a), a) = f_a(x^*(a), a).$$

The indirect effect through  $x^*$  vanishes because at the optimum the objective is flat in  $x$  (the FOC). Intuitively: you are already choosing  $x$  optimally, so a small change in  $a$  does not make re-optimizing  $x$  worth anything to first order.

**Constrained version (for later).** For  $\max_x f(x, a)$  subject to  $g(x, a) = 0$  with Lagrangian  $\mathcal{L} = f(x, a) + \lambda g(x, a)$ , the same logic gives  $V'(a) = \left. \frac{\partial \mathcal{L}}{\partial a} \right|_{\text{optimum}} = f_a + \lambda g_a$ . This is where the multiplier  $\lambda$  earns its “shadow price” interpretation.

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## 2 Preferences

Let  $X$  be the set of consumption bundles and  $\succsim$  a **weak preference** relation ( $x \succsim y$  means “ $x$  is at least as good as  $y$ ”);  $\succ$  denotes strict preference.

- $\succsim$  is **complete** if for all  $x, y \in X$ ,  $x \succsim y$  or  $y \succsim x$  (or both).
- $\succsim$  is **transitive** if for all  $x, y, z$ ,  $x \succsim y$  and  $y \succsim z$  imply  $x \succsim z$ .
- $\succsim$  is **rational** if it is complete and transitive.
- A rational  $\succsim$  is **continuous** if for all  $x, y$  with  $x \succ y$  there is  $\epsilon > 0$  such that any  $x', y'$  within  $\epsilon$  of  $x, y$  (respectively) still satisfy  $x' \succ y'$ . (Small perturbations don't flip strict preferences.)
- $\succsim$  is **reflexive** if  $x \succsim x$  for every  $x$ .
- $\succsim$  is **monotonic** if  $x \geq y$  implies  $x \succsim y$  (“more is at least as good”).
- $\succsim$  is **strictly monotonic** if  $x \geq y$  with  $x_j > y_j$  for some  $j$  implies  $x \succ y$ .
- $\succsim$  is **convex** if for all  $x, y, z$  with  $x \succsim z$  and  $y \succsim z$ , we have  $tx + (1 - t)y \succsim z$  for all  $t \in [0, 1]$  (“averages are at least as good as extremes”).

## 3 Utility Functions

**Definition.** A utility function  $u : X \rightarrow \mathbb{R}$  *represents*  $\succsim$  if for every pair  $x, y \in X$ ,

$$x \succsim y \iff u(x) \geq u(y).$$

- **Existence.** If  $\succsim$  is rational and continuous on  $X$ , there exists a continuous utility function representing it.
- **Non-uniqueness (ordinality).** If  $u$  represents  $\succsim$  and  $\varphi$  is a strictly increasing function, then  $\varphi(u)$  represents the *same*  $\succsim$ . Only the **ordinal** information in  $u$  (the ranking) matters, not its cardinal values.

## 4 Indifference Curves and the MRS

Stick to two goods  $x$  and  $y$ . An indifference curve is the set of bundles with a fixed utility  $c$ :

$$u(x, y) = c \iff u(x, y) - c = 0.$$

Treat this as one equation defining  $y$  implicitly as a function of  $x$  (set  $h(y; x) = u(x, y) - c$ ). By the **IFT**, if  $\partial u / \partial y \neq 0$  then  $y = y(x)$  along the curve and

$$\frac{dy}{dx} = -\frac{\partial u / \partial x}{\partial u / \partial y}.$$

The **marginal rate of substitution** is the negative of this slope,

$$\text{MRS} = -\frac{dy}{dx} = \frac{\partial u / \partial x}{\partial u / \partial y} = \frac{u_x}{u_y},$$

i.e. how much  $y$  the consumer would give up for one more unit of  $x$  while staying on the same indifference curve.

## 5 Exercises

### Exercise 1 (Envelope Theorem)

Let  $V(a) = \max_x \left( ax - \frac{1}{2}x^2 \right)$ , with  $a > 0$ .

- (a) Find  $x^*(a)$  and  $V(a)$  directly.
- (b) Compute  $V'(a)$  from your closed form, and verify it equals  $\frac{\partial}{\partial a} \left( ax - \frac{1}{2}x^2 \right)$  evaluated at  $x = x^*(a)$ .

### Exercise 2 (Preferences and utility: perfect substitutes)

Two goods, apples ( $x$ ) and oranges ( $y$ ), with  $u(x, y) = 2x + y$ .

- (a) With no apples, how many oranges make the consumer indifferent to the bundle (2 apples, 1 orange)?
- (b) Draw the indifference curves for  $u = 5$  and  $u = 7$ . Do they cross? Can indifference curves ever cross? Why or why not?
- (c) Prove that preferences represented by a utility function are transitive: if  $(x_1, y_1) \succsim (x_2, y_2)$  and  $(x_2, y_2) \succsim (x_3, y_3)$ , then  $(x_1, y_1) \succsim (x_3, y_3)$ .

### Exercise 3 (Cobb–Douglas utility)

Let  $u(x, y) = \alpha \log(x) + \beta \log(y)$  with  $\alpha > 0$ ,  $\beta > 0$ , on  $x, y > 0$ . Recall a preference is **monotonic** when more is (weakly) better.

- (a) Are the preferences represented by  $u$  monotonic? Explain.
- (b) What is the sign of the slope of the indifference curves? Are they upward or downward sloping?
- (c) Consider  $\tilde{u}(x, y) = x^\alpha y^\beta$ . Prove  $u$  and  $\tilde{u}$  represent the same preferences:  $u(x_1, y_1) \geq u(x_2, y_2) \iff \tilde{u}(x_1, y_1) \geq \tilde{u}(x_2, y_2)$ .
- (d) The price of  $x$  is  $p$  and the price of  $y$  is 1; the consumer currently sits at  $(x_0, y_0)$ . What should the consumer do if  $\text{MRS}(x_0, y_0) > p$ ? What if  $\text{MRS}(x_0, y_0) < p$ ?
- (e) Suppose instead  $u(x, y) = 2x + y$ . How does your answer to (d) differ?